Systems Engineering Qualification Program

Input for identifying training service providers

Target group: CurioZ Team
Aim: Getting in contact with service providers

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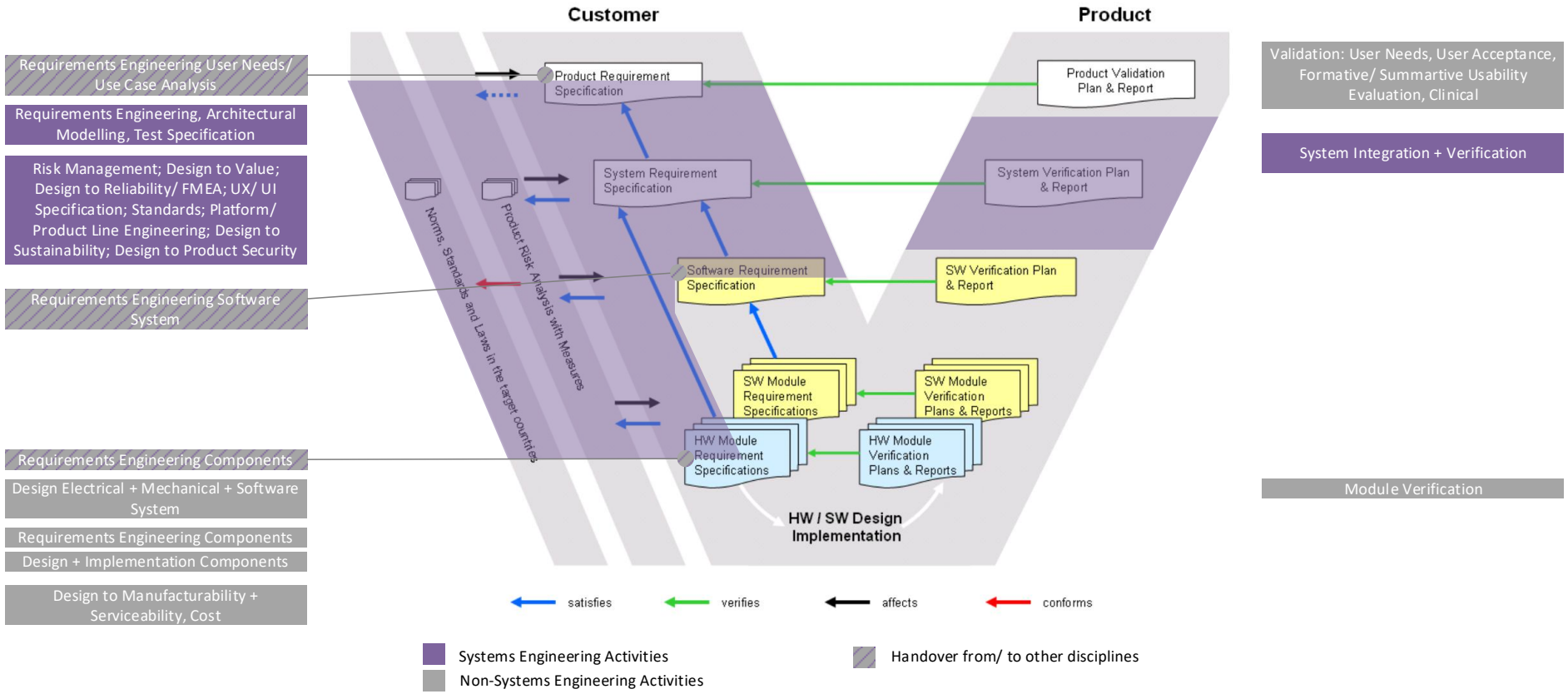
Context and Core Concept

Service Requirements

Contents on Key Competencies

Systems Engineering
Context

Systems Engineering requires all the capabilities needed to successfully work on development projects within the purple area of the V-Model, as well as for concept and advanced development.



Systems Engineering Curriculum
Core Concept

ZEISS Medical Technology (MED) is planning a systematic approach to equip Systems Engineers with the necessary capabilities to excel in their projects. This initiative will strengthen Systems Engineering as a profession at MED and promote standardization across MED locations worldwide. Systems Engineering encompasses multiple aspects, and thus, the curriculum for Systems Engineers at MED will reflect this diversity. The initial focus will be on the interrelated topics of systems architecture, driven by Model-Based Systems Engineering (MBSE) techniques, and traditional requirements engineering.

The curriculum is designed to have levels ranging from A (lowest) to D. Since Systems Engineers are not expected to work in isolation, they need to share a specific set of tools with other engineering disciplines, such as electrical, mechanical, or software engineers. Therefore, some level A training modules will be available for multiple disciplines.

The core personnel to be trained include:

- Systems Engineers: Some of whom are more technical, while others are more focused on medical applications.
- Systems Architects: Who concentrate on either a specific system (medical device/product family/product line) or a networked solution consisting of several devices; lead systems architects who integrate systems within a business sector.

The emphasis on outsourcing training provision is on the lower levels, as there is a significantly larger number of personnel that need to be trained.

Context and Core Concept

Service Requirements

Contents on Key Competencies

Service Description
General

- Concepts for training modules for pre-defined contents; structure/ modality didactically optimized.
- Creation of training material (English).
- Alignment with ZEISS regarding training contents and modalities; review of material.
- Provision of training; the trainings shall include hands-on exercises with the given tools
- Reviewing the correct and goal-driven application of the training participants for certification: SysE.A, SysE.B, SysE.C, SysA.C, SysA.D, AppD.B, AppD.C, SlnA.C

Service Description
Requirements/ Constraints

- Trainer shall be proficient in systems engineering, especially in the topics to be trained as well as in providing professional trainings.
- For some training modules the trainer shall be able to delve into ZEISS MED processes.
- Trainer shall be willing to travel to respective ZEISS locations (Jena, Oberkochen, Dublin, California). The trainings will be on-site.
- Trainer shall be able to provide the training in English.
- For the training, a fictional medical device should be used as a continuous example.
- The size of the training groups will not exceed 12 participants.
- There may be several trainers, but the training material needs to be unified.
- All current development personnel shall be trained with respect to the individual level within 2 years from start of first training
- Tools: Polarion, Enterprise Architect

Service Description
Personnel subject to training

- All Engineers (for basic requirements engineering and architecture only)
- Systems Engineers (incl. Application Developers = Systems Engineers focussed on medical application)
- Architects: System, Software, Solution

Number of persons for locations and training modules (see slides below for denomination of training modules)

		Business Sector																																	
		NES	ETP	DEN	REF	SAS		CDM		SPS, DORC	Adv. OPT	Total			RqE1			RqE2			RqE3			SyA1			SyA2			SyA3			SyA4		
															JEN	OKO	DUB	JEN	OKO	DUB	JEN	OKO	DUB	JEN	OKO	DUB	JEN	OKO	DUB	JEN	OKO	DUB	JEN	OKO	DUB
Target Qualification Level	E.A	140	28	0	36	29	7	67	33	0	3	343			79	197	67							79	197	67									
	SysE.A	2	0	0	1	3	1	1	0	1	0	9			2	6	1							2	6	1									
	SysE.B	6	0	0	10	12	3	3	1	5	0	40			14	23	3	14	23	3				14	23	3	14	23	3						
	SysE.C	3	1	0	5	5	2	1	1	3	0	21			8	12	1	8	12	1	8	12	1	8	12	1	8	12	1						
	SysA.C	2	0	1	2	0	0	0	0	0	0	5			2	3	0	2	3	0	2	3	0	2	3	0	2	3	0	2	3	0			
	SysA.D	1	0	1	0	0	0	0	0	0	0	2			0	2	0	0	2	0	0	2	0	0	2	0	0	2	0	0	2	0			
	AppD.A	0	0	0	0	0	0	1	1	0	0	2			1	0	1							1	0	1									
	AppD.B	0	0	0	4	0	0	6	3	0	0	13			7	0	6	7	0	6				7	0	6	7	0	6						
	AppD.C	0	1	0	3	0	0	3	1	0	0	8			4	1	3	4	1	3	4	1	3	4	1	3	4	1	3						
	SlnA.C	0	0	0	1	0	0	0	0	0	0	1			1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1	0	0	1		
	SwA.C	1	0	1	2	0	0	0	0	0	0	4			2	2	0	2	2	0	2	2	0	2	2	0	2	2	0	2	2	0			
	Sum	155	30	3	64	49	13	82	40	9	3	448			120	246	82	38	43	13	17	20	4	120	246	82	38	43	13	5	7	0	1	0	0
	Location:	OKO	OKO	OKO	JEN	OKO	JEN	DUB	JEN	OKO	JEN				448			94			41			448			94			12			1		

Locations: JEN: Jena | OKO: Oberkochen | DUB: Dublin, CA

Context and Core Concept

Service Requirements

Contents on Key Competencies

Topics
Key Competencies

Key competencies for first wave training:

(1) Requirements Engineering (requirements, design constraints)

(2) Architecture and Modelling (structural/ logical, functional)

Training Module RqE1, Requirements Engineering (RE)

Level A: SysE.A, E.A, SwE.A, AppD.A

Goals

- Understanding the role of RE in system development
- Understanding, how RE is related to architecture and design
- Understanding the rules for requirements (properties, text templates)

Contents

1. Basics: Definition/ Purpose of requirements; V-Model;; principles of requirements engineering; system and context boundary; Characteristics/ quality criteria of requirements
2. Problem Domain + Solution Domain. Requirements and design specification items/ architecture; Basic architecture; glossary;
3. Regulatory and statutory requirements;
4. Types of requirements: Functional; quality; technical; user interface; others;
5. Text templates
6. Attributes for requirements
7. Outlook: Special analyses (DtV, DfR, ...) and requirements, AI

Training Module RqE2, Requirements Engineering (RE)

Level B: SysE.B, AppD.B

Goals

- Understanding the relations to standard MED documents
- Understanding the levels and types of requirements specifications
- Understanding the Zig-Zag model
- Understanding the relation semantics
- Understanding the relationship to important MED processes (e.g. RM, UE, Sw)

Training Module RqE2, Requirements Engineering (RE), cont‘d

Level B: SysE.B, AppD.B

Contents

1. Recap
2. levels of requirements + architecture, Zig-Zag model
3. Requirements Documents : (PRS, SRS, SDS, CRS, CDS, SWRS, dSWS, WFRS)
4. Relations: traceabiliy;
5. Requirements in Development Stages: Requirements and architecture; ; advance development; product development; lifecycle;
6. RE within the product generation processes: PDP, RM, UE, SWDP
7. Internal Requirements: Manual, Assembly, Service; Platform Components (Standard descriptions, reuse, linking); Variants

Training Module RqE3, Requirements Engineering (RE)

Level C: SysE.C, SysA.C, SwA.C, AppD.C

Goals

- Understanding the interrelation of requirements and architecture
- Unterstanding elicitation techniques for requirements
- Understanding how to define interfaces
- Procedure for requirements engineering, architecture

Contents

1. Recap
2. Requirements and Modelling
3. Requirements in Development Stages: Requirements and architecture;
4. Determination of requirements: sources of requirements;
5. Techniques for identifiying/ eliciting requirements (I); e.g. voice of customer, SQuaRE; FMEA; HAZOP;
6. Techniques for identifiying/ eliciting requirements (II): Based on architectural views (context, behavioral) and mental models (?)
7. deriving parameters by deduction; application domain to quality
8. Requirements Engineering as a process; classical, agile; iterative
9. Knowledge Management

Training Module SyA1, Systems Architecture
Level A: SysE.A, E.A, SwE.A, AppD.A

Goals

- Understanding what is an architecture and what are the key aspects
- Understanding what is modelling and its relation to architecture
- Understanding how to read/ interpret architectural descriptions in SysML 1.X.
- Understanding how architectural descriptions/ models are related to requirements;

Training Module SyA1, Systems Architecture, cont'd

Level A: SysE.A, E.A, SwE.A, AppD.A

Contents

1. Introduction and terminology
2. Architecture Levels and architectural views
3. Requirements and Architecture: The Zig-Zag Pattern
4. Reading models
5. Use Case Models
   - Reading use case diagrams (actors, use cases, relations, ...);
   - Stakeholder and Context Models
   - Reading Block Definition Diagrams (types, instances, attributes methods/ functions, ...); stakeholder model, context model;
   - Use Case Activity Models
   - Reading Activity diagrams (activities, actions, ...)
   - Logical Architecture Modells
   - Revisiting Block Definition Diagrams, reading Internal Block Diagrams (objects, ports, ...);
   - Additional behavioral models; Sequences, state machines;

Training Module SyA2, Systems Architecture

Level B: SysE.B, AppD.B

Goals

- Understanding how to work with the Modelling Tool to read architectural descriptions
- Understand to read versions and variants
- Understand the reference model in MED

Contents

- Recap
- Introduction to the Enterprise Architect
- Introduction to the global structure of MED models
- Creating and updating architectural models
- Introduction to versioning, comparison, baselining

Training Module SyA3, Systems Architecture
Level C: SysA.C, SlnA.C, SwA.C

Goals

- Understand the subset of SysML 1.X notation to be used @MED and its semantics
- Understanding for which point of view which SysML element are needed
- Understanding how to use modelling within which phase of project
- Understanding how to define interfaces
- Understanding how the model/ the architecture is developed aligned with requiremenst (procedure)
- Understand how the models can support special topics, e.g. safety, usability, reliability
- Understand how the models can provide orientation for the development team

Training Module SyA3, Systems Architecture, cont'd

Level C: SysA.C, SlnA.C, SwA.C

Contents

1. SysML notational elements to be used @MED for what view/ purpose
2. Block definition diagram
3. Internal block diagram
4. Use case diagram
5. Activity diagram
6. Sequence diagram
7. State machine diagram
8. Requirements diagram
9. Integrating based on reference model
10. Views for several disciplines in systems development

Training Module SyA4, Systems Architecture
Level C: SlnA.C

Goals

- Understands how to model functional connectivity
- Understand how to do configuration and version management in a solution
- Understands how to assign functions/ quality properties (e.g. risk) among the elements of a solution
- Understand how to optimize the solution for regulatory aspects

Contents

1. Recap
2. Modelling device configuration and versions
3. Modelling user workflows
4. Modelling data flows
5. Modelling safety and regulatory criticality

Low prio – currently only one person

ZEISS

Seeing beyond